

$$\begin{aligned} 1. \quad Q &= CV \\ Q &= 4(12) \\ Q &= 48.0 \text{ C} \end{aligned}$$

$$\begin{aligned} 2. \quad C &= \frac{Q}{V} \\ C &= \frac{10\mu}{10} \\ C &= 1\mu F \end{aligned}$$

$$3. \quad C_2 = 12\mu \text{ \& } C_3 = 24\mu \text{ are in series}$$

$$\begin{aligned} C_{23} &= \left(\frac{1}{12\mu} + \frac{1}{24\mu} \right)^{-1} \\ C_{23} &= 8\mu F \end{aligned}$$

$$\begin{aligned} C_{234} &= C_{23} + C_4 \\ C_{234} &= 8\mu + 12\mu \\ C_{234} &= 20\mu F \end{aligned}$$

$$\begin{aligned} C_{eff} &= \left(\frac{1}{20\mu} + \frac{1}{20\mu} \right)^{-1} \\ C_{23} &= 10\mu F \end{aligned}$$

$$\begin{aligned} 4. \quad C_{12} &= C_1 + C_2 \\ C_{12} &= 20\mu + 10\mu \\ C_{12} &= 30\mu F \end{aligned}$$

$$\begin{aligned} C_{34} &= C_3 + C_4 \\ C_{34} &= 30\mu + 30\mu \\ C_{34} &= 60\mu F \end{aligned}$$

$$\begin{aligned} C_{eff} &= \left(\frac{1}{30\mu} + \frac{1}{60\mu} \right)^{-1} \\ C_{23} &= 20\mu F \end{aligned}$$

$$\begin{aligned} 5. \quad C_{12} &= \left(\frac{1}{15\mu} + \frac{1}{3\mu} \right)^{-1} \\ C_{12} &= 2.5\mu F \end{aligned}$$

$$\begin{aligned} C_{123} &= 2.5\mu + 6\mu \\ C_{123} &= 8.5\mu F \end{aligned}$$

$$\begin{aligned} C_{eff} &= \left(\frac{1}{8.5\mu} + \frac{1}{20\mu} \right)^{-1} \\ C_{eff} &= 5.96\mu F \end{aligned}$$

$$6. C_{eff} = \left(\frac{1}{7\mu} + \frac{1}{5\mu} \right)^{-1} + 4\mu + 6\mu$$

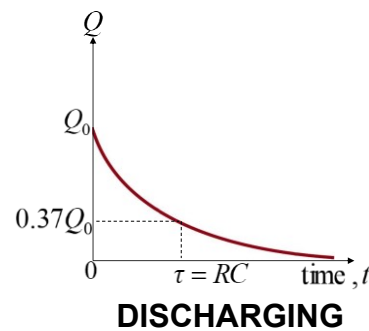
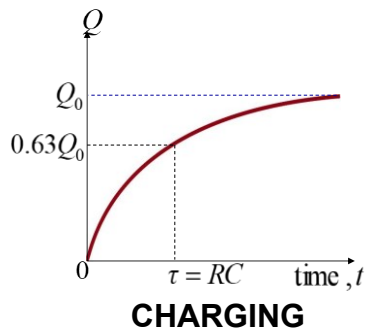
$$C_{eff} = 12.92\mu F$$

$$7. \quad \tau = RC$$

$$\tau = (50 \times 10^3)(20 \times 10^{-6})$$

$$\tau = 1.0s$$

8.



$$9. \quad i)$$

$$\tau = RC$$

$$\tau = (20 \times 10^3)(12 \times 10^{-6})$$

$$\tau = 0.24s$$

ii)

$$Q = Q_0 e^{-\frac{t}{RC}}$$

$$Q = 72 \times 10^{-6} \left(e^{-\frac{0.5}{0.24}} \right)$$

$$Q = 8.97 \times 10^{-6} C$$

$$Q = CV = (12\mu)(6) = 72\mu C$$

$$10. \quad Q_0 = CV_0$$

$$Q_0 = (1000 \times 10^{-6})(4)$$

$$Q_0 = 4.0 \times 10^{-3} C$$

$$Q = Q_0 \left(1 - e^{-\frac{t}{RC}} \right)$$

$$Q = 4 \times 10^{-3} \left(1 - e^{-\frac{5}{(2 \times 10^3)(1000 \times 10^{-6})}} \right)$$

$$Q = 3.67 \times 10^{-3} C$$

